A complete all-smooth F5 control

For $y^2 = x^5 - x$ over $\mathbf{F}_5$, Lean proves there are exactly 16 actual reduced polynomial pairs. Every u splits and every $v = 0$. The curve polynomial is monic, degree five and separable. Identity is included. This supplies the fixed all-smooth native control before data. [pcc:unverified]

Exact control	Complete pairs	Split u	Smooth fraction
Same equation over F3	8	7	$7/8$
Same equation over F5	16	16	1

These are exact polynomial-pair models, with complete coefficient-shape proofs. The formal identification with the geometric Picard/Jacobian group remains separate. Two small fields do not refute a fixed-genus asymptotic limit or a qualified large-prime/typical-subgroup heuristic. [pcc:unverified]

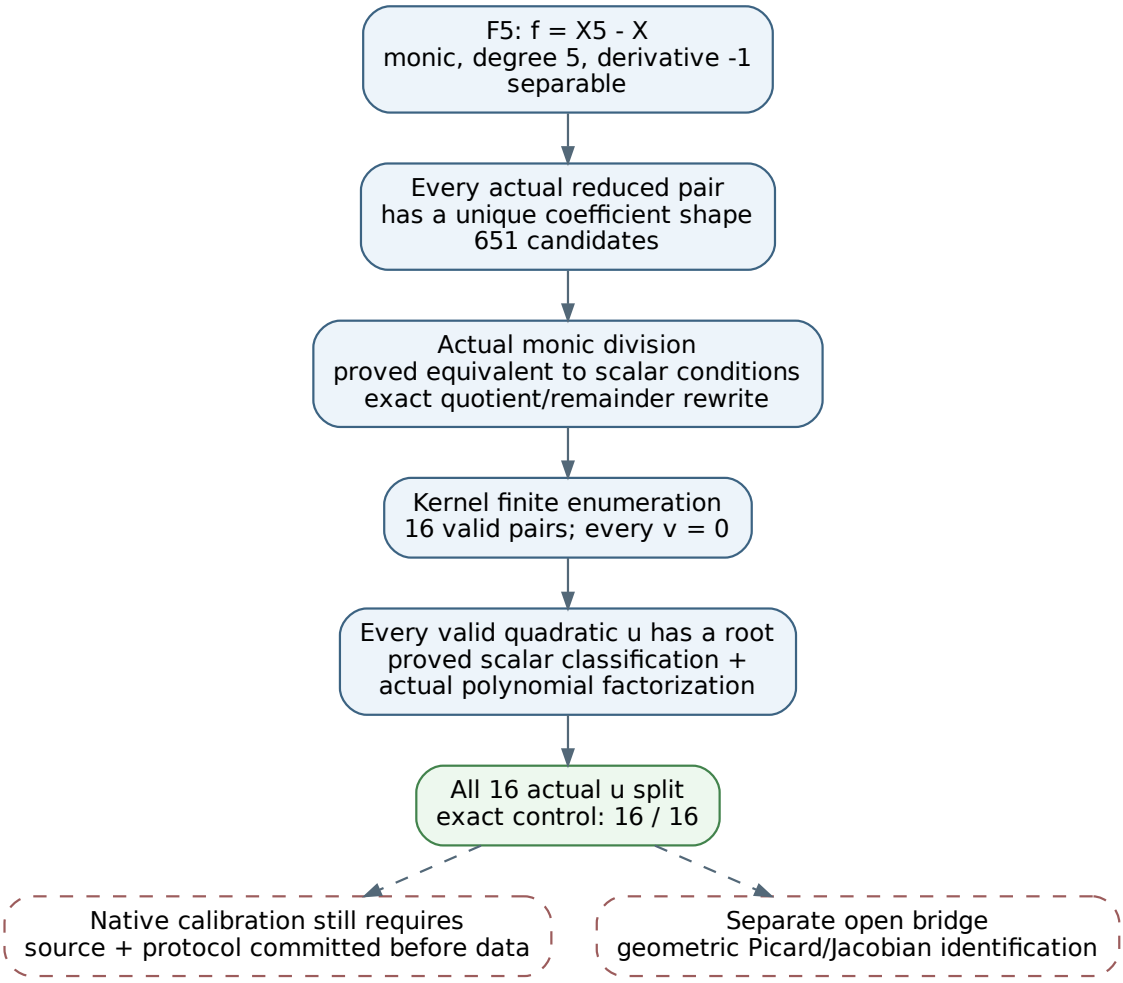


Figure 1: Actual polynomial completeness and splitting give a pre-data calibration control. Dashed obligations remain open.

Proof and execution scope

The model includes every monic u with $\deg u \leq 2$, every v with $\deg v < \deg u$, and requires $u \mid v^2 - f$. An explicit bijection with the complete coefficient shapes proves the cardinality. Distinct shapes give distinct polynomial pairs. [pcc:unverified]

An exact monic quotient/remainder identity proves the actual division test equivalent to finite scalar equations. Kernel `decide` classifies 651 candidates and finds 16 valid pairs. Every valid quadratic u has a root r over F5. The actual factorization $(X-r)(X+a+r) = X^2 + aX - r^2 - ar$ proves splitting. This is the `Polynomial.Splits` predicate. [pcc:unverified]

The curve derivative is -1, giving separability by an explicit Bezout certificate. Scalar v-zero conditions are proved for every candidate and transferred to every actual pair. The proof adapts the earlier F3 shape method while rechecking characteristic-five equations. A recursion bound 4096 is a finite-reduction resource limit, with no trust relaxation. [pcc:unverified]

The combined binding is:

`MumfordFiveControlAudit.mumford_five_control_certificate`.

The whole tracked project gate passes 240 declarations, 0 failures and 0 trusted-native exceptions, including generated declarations. Only propext, Classical.choice and Quot.sound are accepted. Lean 4.35.0 and Mathlib revision `56a005855ea17e0b0958f73ec801ae140b0a1679` are pinned. The failed first build and passing second build are retained. Lake's module-system advisories remain visible. [pcc:unverified]

Source commit `11100a85` precedes packet `PCC-20261109-e81718` and scalar supplement `IDC1ha387b9e22b14b9cf`. The full local packet check passes at 21:56:51Z on producer host `762f5fc120c9`; signed standing remains unverified. The supplement accepts 32 derived and 11 extra seeded points over $2^{61} - 1$ for the supplied-root quadratic factorization only: three variables, degree two. It does not prove the cardinality, completeness, splitting or geometry. The finite control rests on Lean. [pcc:unverified]

The reused native Rust driver and separate C++ arithmetic implementation build successfully. They have not been invoked on curves. Calibration protocol f2572b4d was frozen at 21:52:34Z and must be committed before affected calls. It requires exact F3/F5 counts and unequal/equal detector controls, all actual addition/group/subgroup checks, source and binary pins, failure retention and no paid compute. Its elapsed intervals are scheduling diagnostics, with no scientific timing claim. [pcc:unverified]

Stoll2014's representation/addition sections remain read at the scopes in ../../cantor-sources.json. Example 4.19 (pp. 25–26) was also read in full for a source-known nonzero-v/order-five control of the later full F3 census. That is a pre-data prediction and still needs a faithful Lean extension. Cantor1987's original publisher PDF remains unread after 403. No source is promoted merely because a related note was read. [pcc:unverified]

Exact source/binding hashes and reproduction are in ../../FIVE_CONTROL.md and ../../five-control-sources.json. The original fourteen-item audit remains partial. Geometry, scaling families, qualified asymptotics, independent-machine validation and the other original bounds remain open. No performance or curve-comparison metric changed. [pcc:unverified]